

KMtracker User Manual

Mileage, fuel, expenses and maintenance for all your vehicles, in your own home.

This guide walks you through KMtracker, screen by screen. You don't need any technical knowledge: if you can use a website, you can use KMtracker. To install it, see the installation section of the README or the Installation and configuration chapter of this manual.

> **Language:** this manual is also available in Spanish (`manual-es.md`).

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1. What is KMtracker

KMtracker is a **web application** that runs on your own machine (a computer, a mini-PC, a Raspberry Pi or a NAS). It helps you keep track of your cars and motorcycles: the kilometres you drive and what you spend on fuel, repairs and maintenance.

Three simple ideas:

- **Your data is yours.** It is stored on your machine, in a single file. Nothing is sent to external services.
- **You use it from your browser** on a phone, tablet or computer. There is nothing to install on your devices.
- **Optionally**, you can log entries from your phone with a **Telegram bot**.

Every menu of the application is always in the top bar:

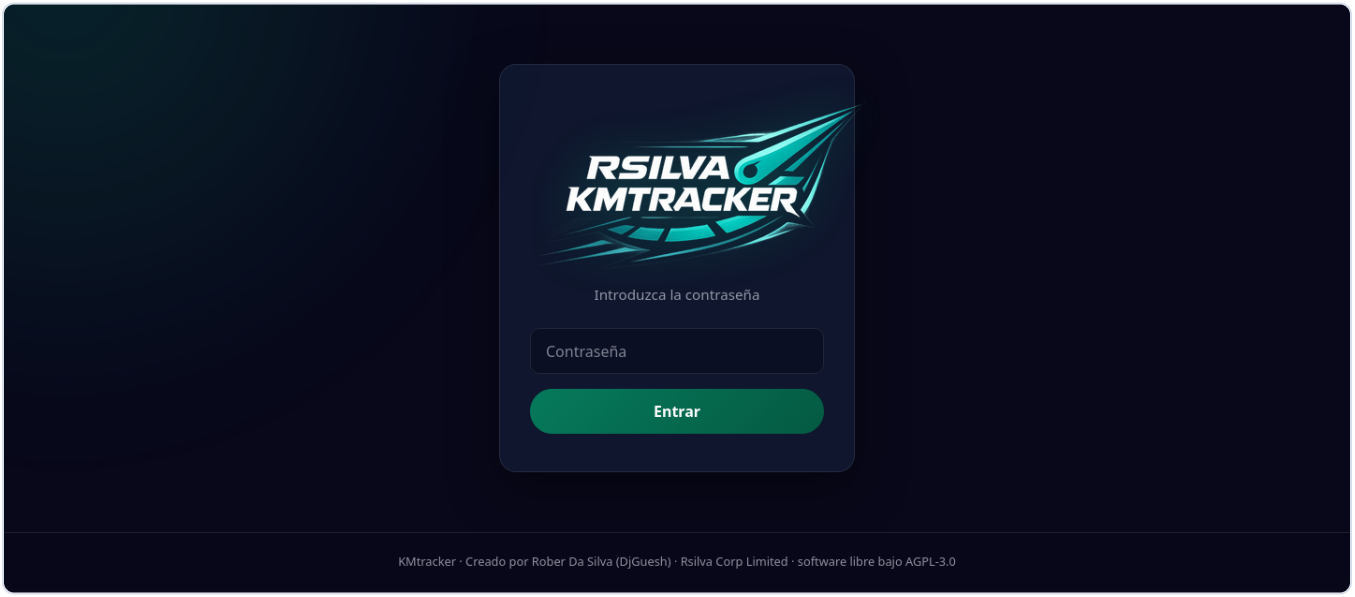
Panel · Registrar · KM · Combustible · Mantenimiento · Coches · Informes · Historial · Ajustes · Admin

> The user interface is in Spanish. This manual lists each screen by its English name and, where useful, its Spanish menu label in brackets.

2. Getting started

Signing in

Open your KMtracker address in the browser (for example `http://localhost:5050`) and enter the password defined during installation.

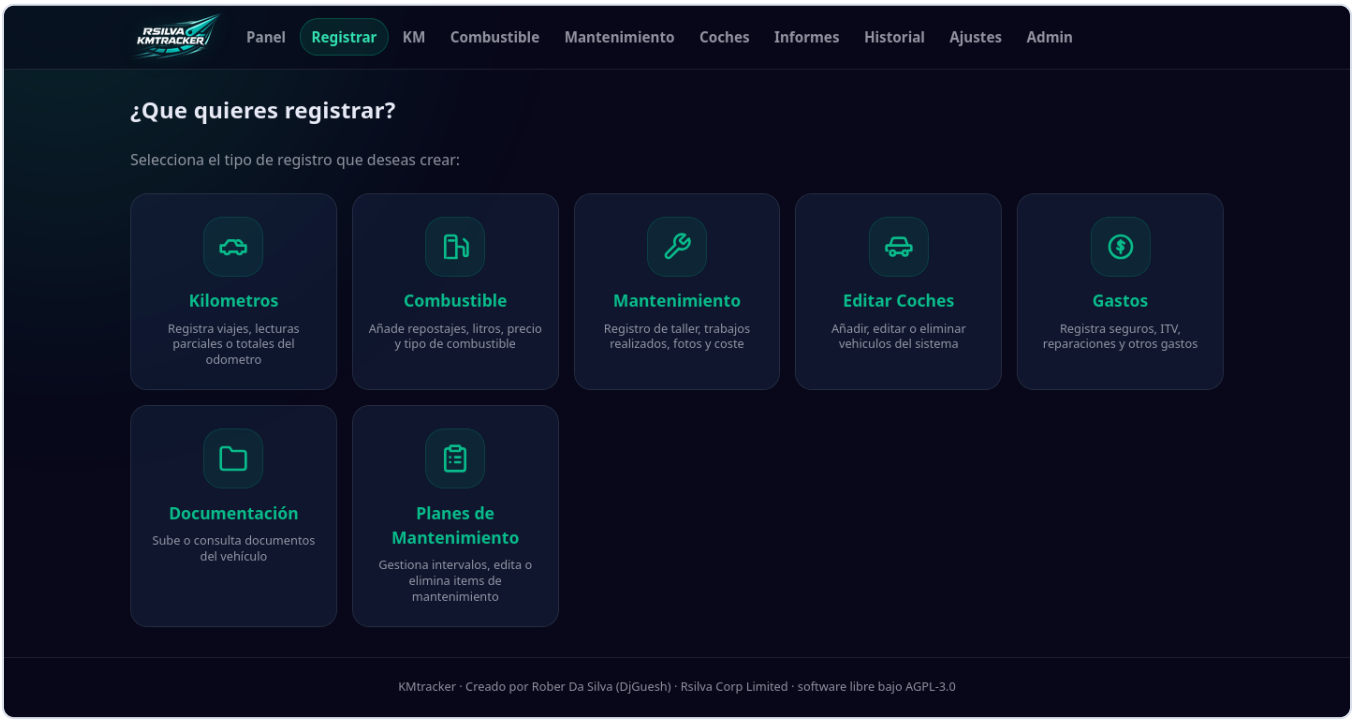


- There is a **single** password for the whole installation (no users or accounts). It is designed for personal or family use.

- After several failed attempts, the application temporarily blocks new attempts to prevent brute-force attacks.
- To sign out, use the **logout** link in the top-right corner.

The "Registrar" menu

The **Registrar** (Log) button is a shortcut that gathers the four most common day-to-day actions:



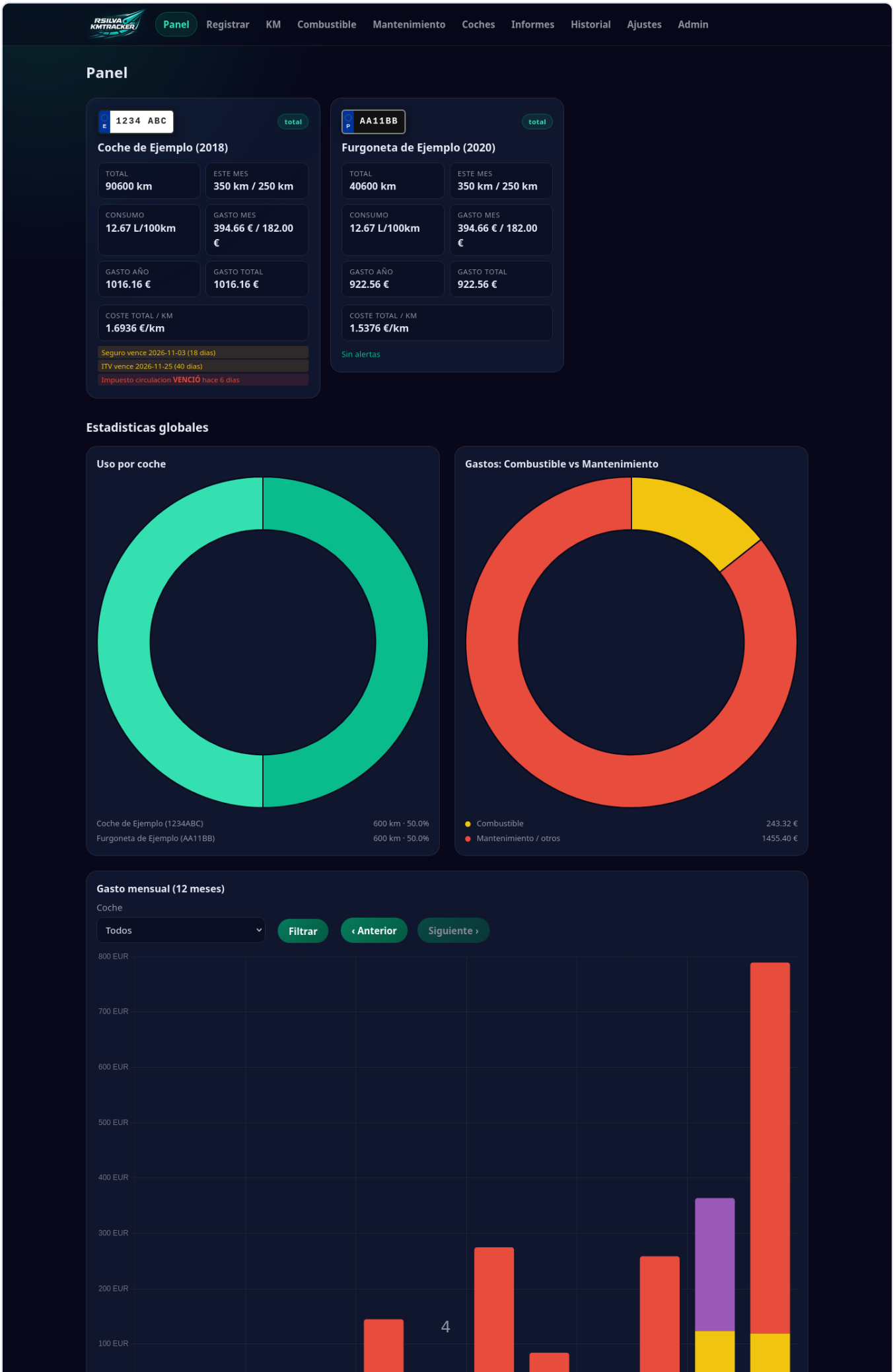
- **Kilómetros** (Mileage) — add a trip or an odometer reading.
- **Combustible** (Fuel) — add a refuelling.
- **Gastos** (Expenses) — add an expense.
- **Documentos** (Documents) — add a document.

Each of these is also available from its own menu in the top bar.

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3. The dashboard

The **Panel** (Dashboard) is the main screen and the summary of your whole fleet. It opens automatically when you sign in.


Panel
Registrar
KM
Combustible
Mantenimiento
Coches
Informes
Historial
Ajustes
Admin
Panel
1234 ABC
total
Coche de Ejemplo (2018)
TOTAL
90600 km
ESTE MES
350 km / 250 km
CONSUMO
12.67 L/100km
GASTO MES
394.66 € / 182.00 €
GASTO AÑO
1016.16 €
GASTO TOTAL
1016.16 €
COSTE TOTAL / KM
1.6936 €/km
Seguro vence 2026-11-03 (18 dias)
ITV vence 2026-11-25 (40 dias)
Impuesto circulacion VENCIÓ hace 6 dias
AA11BB
total
Furgoneta de Ejemplo (2020)
TOTAL
40600 km
ESTE MES
350 km / 250 km
CONSUMO
12.67 L/100km
GASTO MES
394.66 € / 182.00 €
GASTO AÑO
922.56 €
GASTO TOTAL
922.56 €
COSTE TOTAL / KM
1.5376 €/km
Sin alertas
Estadisticas globales
Uso por coche
Coche de Ejemplo (1234ABC) 600 km · 50.0%
Furgoneta de Ejemplo (AA11BB) 600 km · 50.0%
Gastos: Combustible vs Mantenimiento
Combustible 243.32 €
Mantenimiento / otros 1455.40 €
Gasto mensual (12 meses)
Coche
Todos
Filtrar
‹ Anterior
Siguiente ›
800 EUR
700 EUR
600 EUR
500 EUR
400 EUR
300 EUR
200 EUR
100 EUR

Per-vehicle cards

Each car or motorcycle appears in a card with its current figures:

- **Total** — accumulated kilometres of the vehicle.
- **Este mes** (This month) — kilometres driven this month, compared with the previous month.
- **Consumo** — real consumption in **L/100 km**.
- **Gasto mes** — this month's spending, with the previous month as reference.
- **Gasto año** — spending so far this year.
- **Gasto total** — everything spent since you added the vehicle.
- **Coste total / km** — the real cost per kilometre.

If a vehicle has active warnings (upcoming maintenance, insurance, roadworthiness test or road tax), they are highlighted at the bottom of its card.

Global statistics

Below the cards, the dashboard shows charts for the whole fleet:

- **Uso por coche** (Usage per car) — the split of kilometres between vehicles.
- **Gastos: combustible vs mantenimiento** — where the money goes.

Monthly spending (12 months)

A bar chart of the last twelve months' spending, split by fuel, expenses and maintenance. You can filter by vehicle and move back and forward in time with the **Filtrar**, **← Anterior** and **Siguiente →** buttons.

Monthly statistics

The lower block lets you pick a **car**, a **year** and a **month** and see:

- the **daily spending** for that month,
- the **spending per car** for that month.

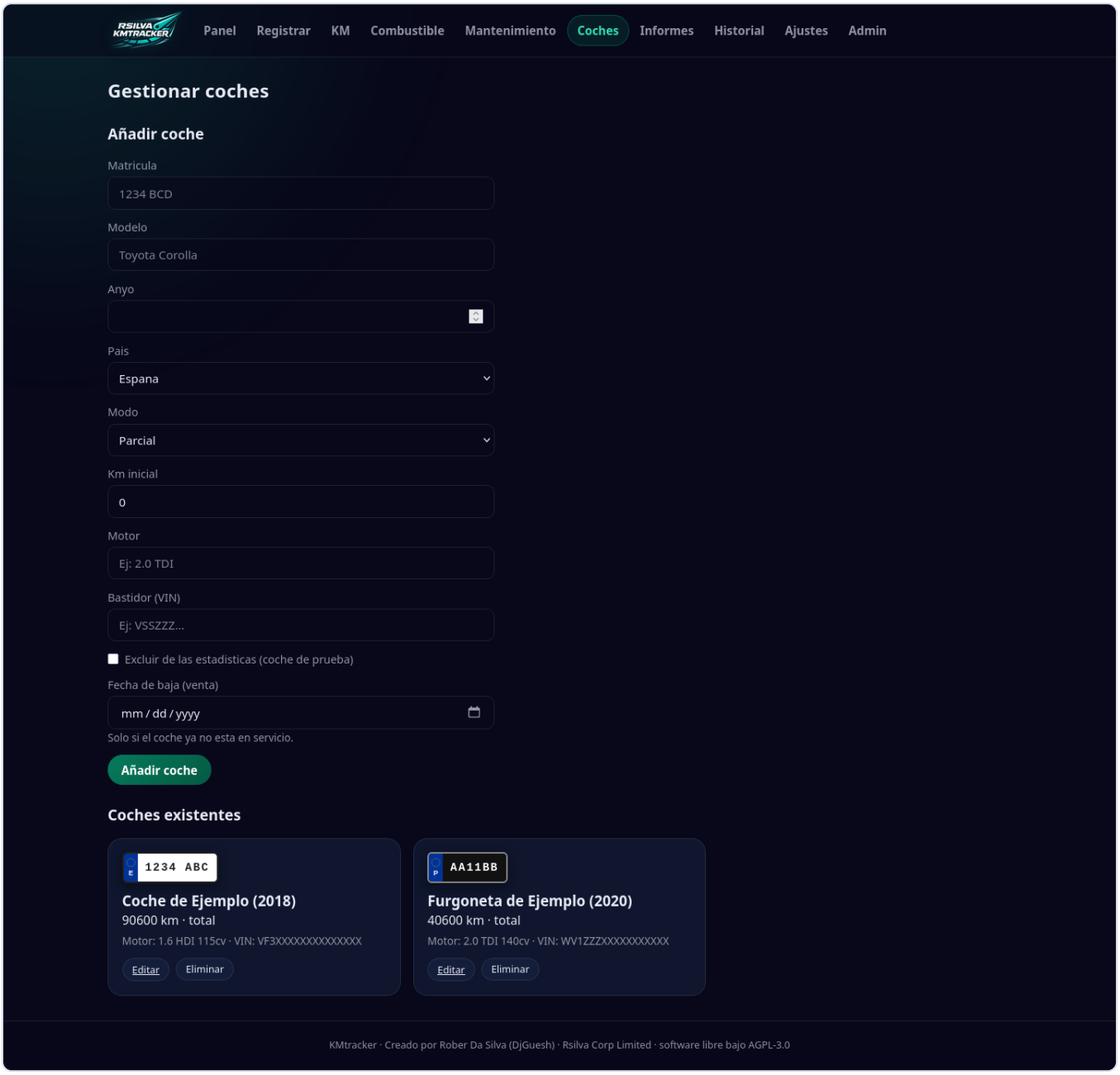
> Which cards and blocks appear on the dashboard is chosen in Settings.

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4. Vehicles

Adding a vehicle

Open **Coches** (Cars), where you'll find the list of vehicles and the form to add a new one.

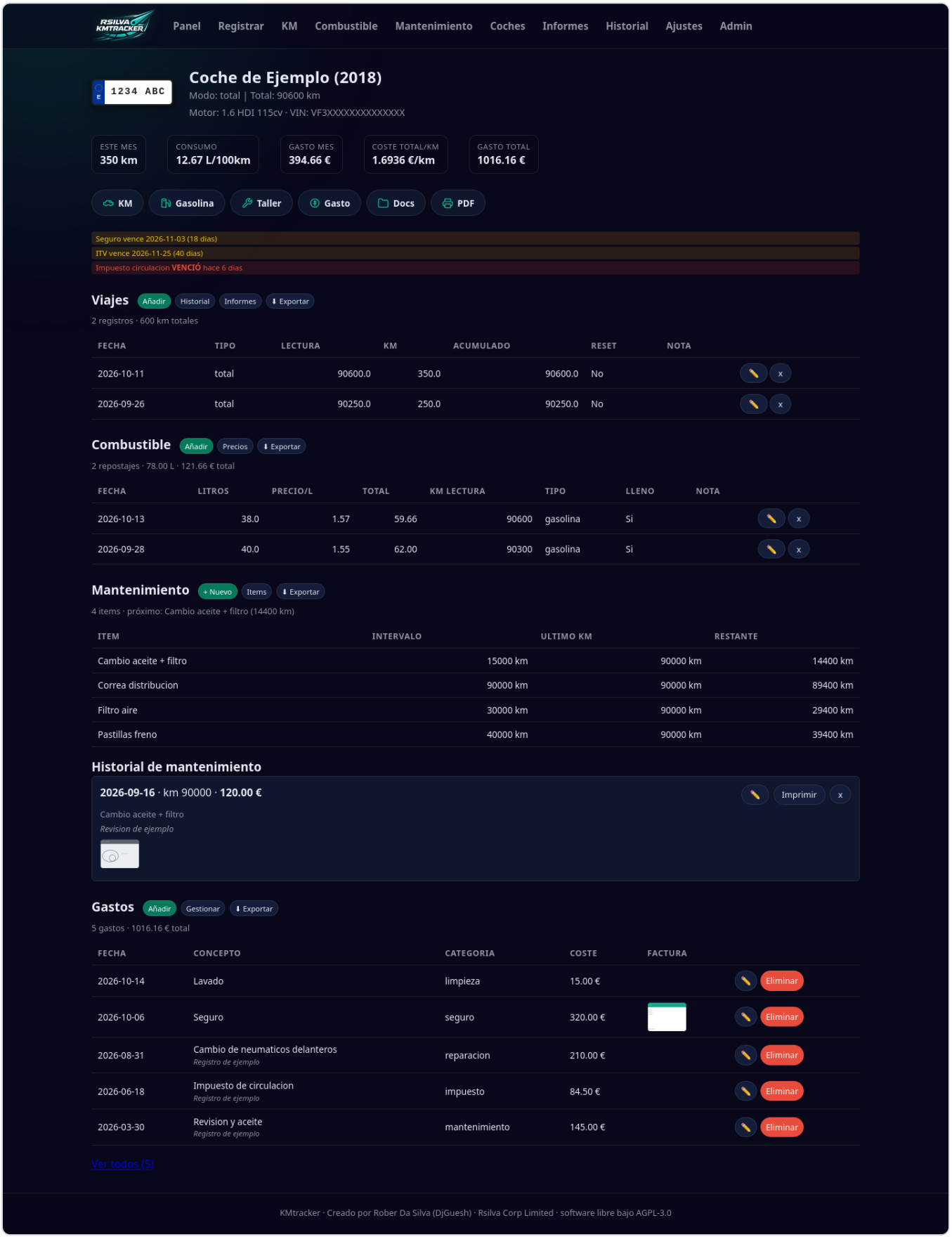


When adding a vehicle, you are asked for:

- **Matrícula** (Plate) — validated for **Spanish and Portuguese** plates.
- **Modelo** — for example, "Seat Leon".
- **Año** (Year) — registration year.
- **Mileage mode** — how you measure kilometres:
- **Total**: you use the odometer that never resets.
- **Parcial**: you use the trip meter that is reset to zero.
- **Initial kilometres** — the reading you start tracking from.
- **Photo** (optional) — a picture of the vehicle.

The vehicle detail page

Click a vehicle to open its **detail page**, the control centre for that specific car.



At the top you'll find its data, the summary indicators (this month, consumption, monthly spend, cost/km, total spend) and the **renewal warnings**. The quick action buttons (**KM**, **Gasolina**, **Taller**, **Gasto**, **Docs**, **PDF**) take you straight to that vehicle's corresponding form.

Below are its sections, each with its own rows and **edit** (pencil) and **delete** (cross) buttons:

- **Viajes / Mileage**
- **Fuel**
- **Maintenance** (with the tasks, their next due point and remaining km)
- **Maintenance history**
- **Expenses**

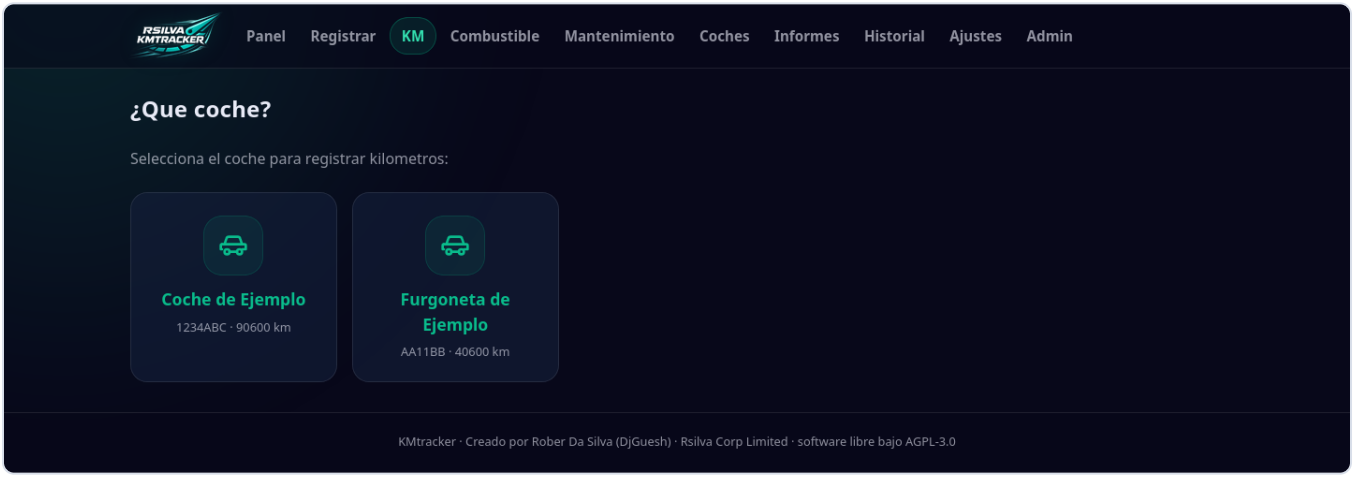
Marking a vehicle as "test" or "sold"

From the detail page you can edit the vehicle to mark it as **test** or as **sold/decommissioned**. An archived vehicle is **not deleted**: it keeps its whole history, it just no longer appears as active.

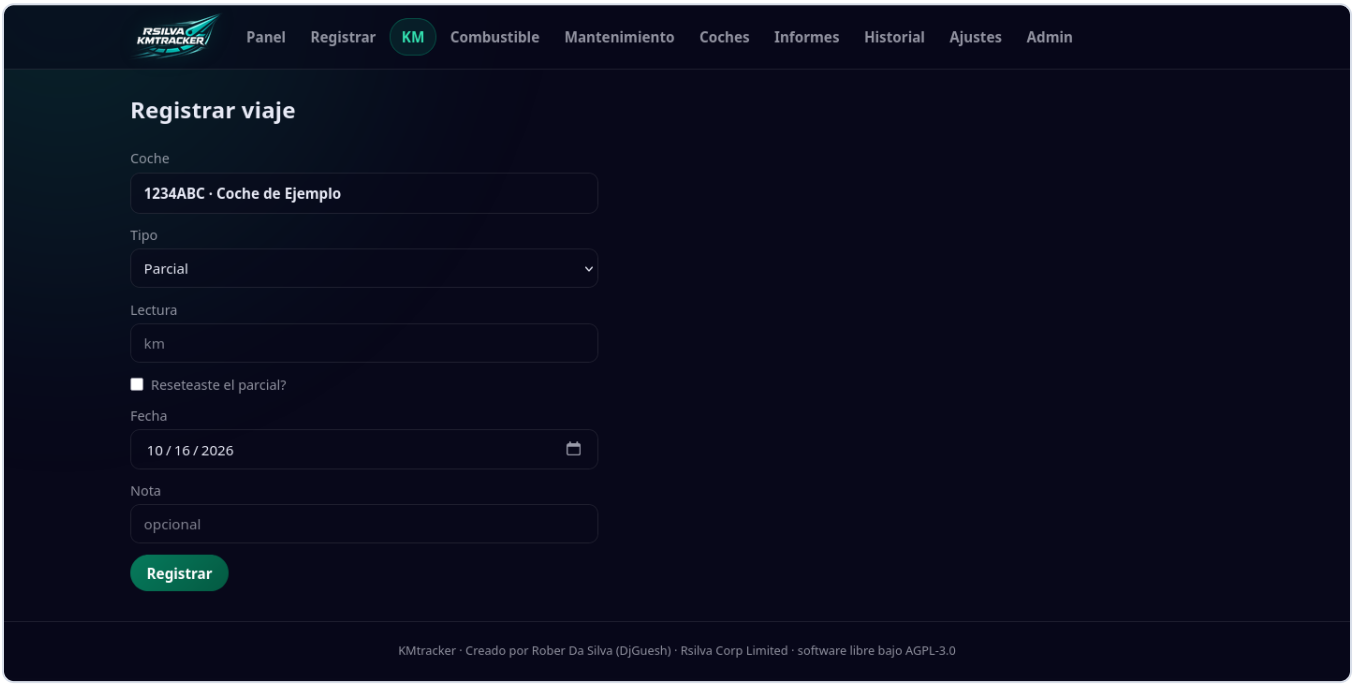
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5. Mileage

In **KM** you record the kilometres driven. First choose the vehicle:



Then fill in the form:



Depending on how the vehicle is configured:

- **Total odometer** — you enter the **odometer reading** (for example `90600`) and KMtracker computes the distance since the previous reading.
- **Trip meter** — you enter the **trip kilometres**.

The result is computed automatically; you don't do any maths.

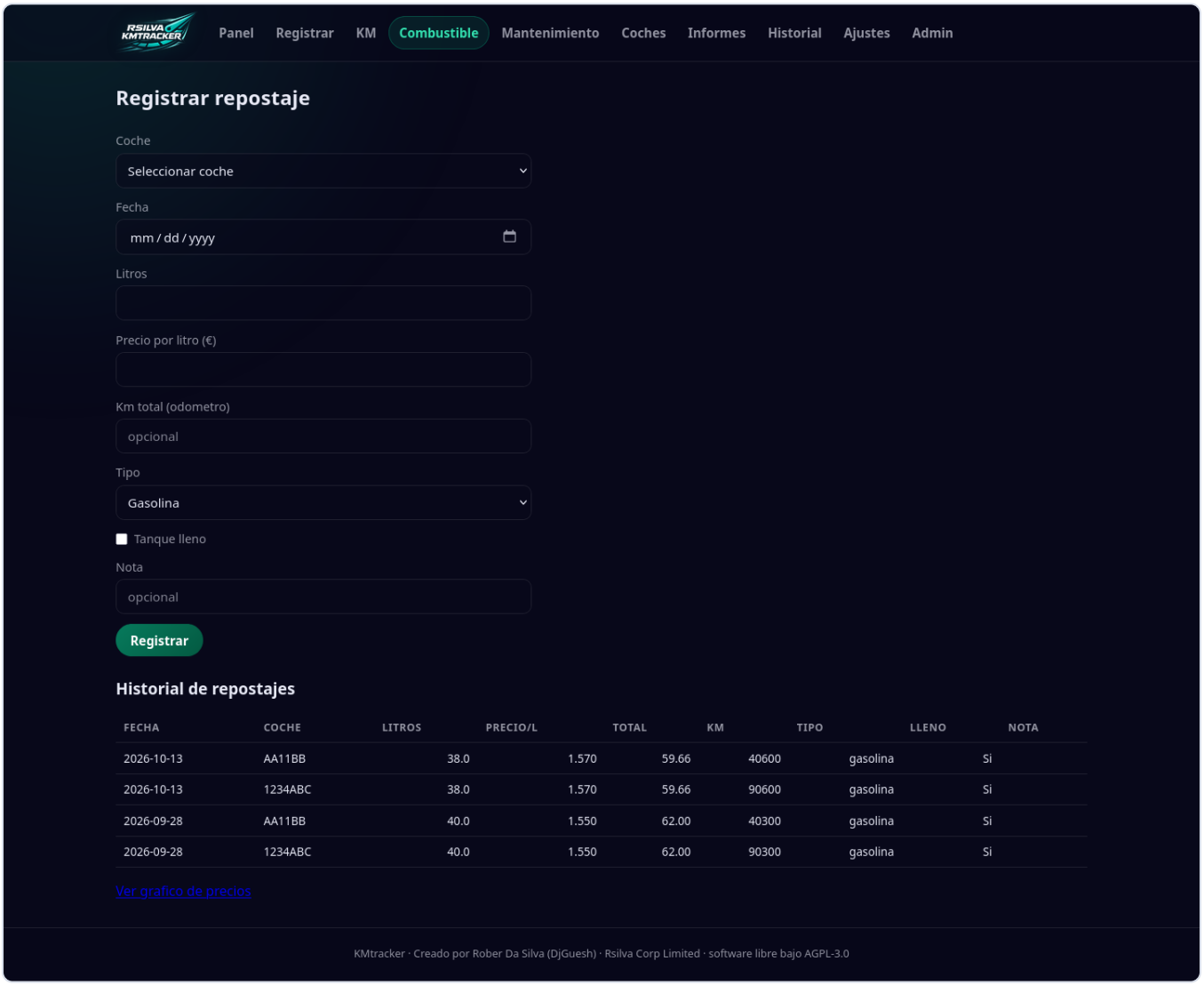
Fixing a mistake

If you typed something wrong, click the **pencil** on the row and correct the value. KMtracker **recalculates in a chain** every affected distance and consumption figure. The full mileage history of each vehicle is always available on its detail page.

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6. Fuel

In **Combustible** (Fuel) you record refuellings and the real consumption is computed.



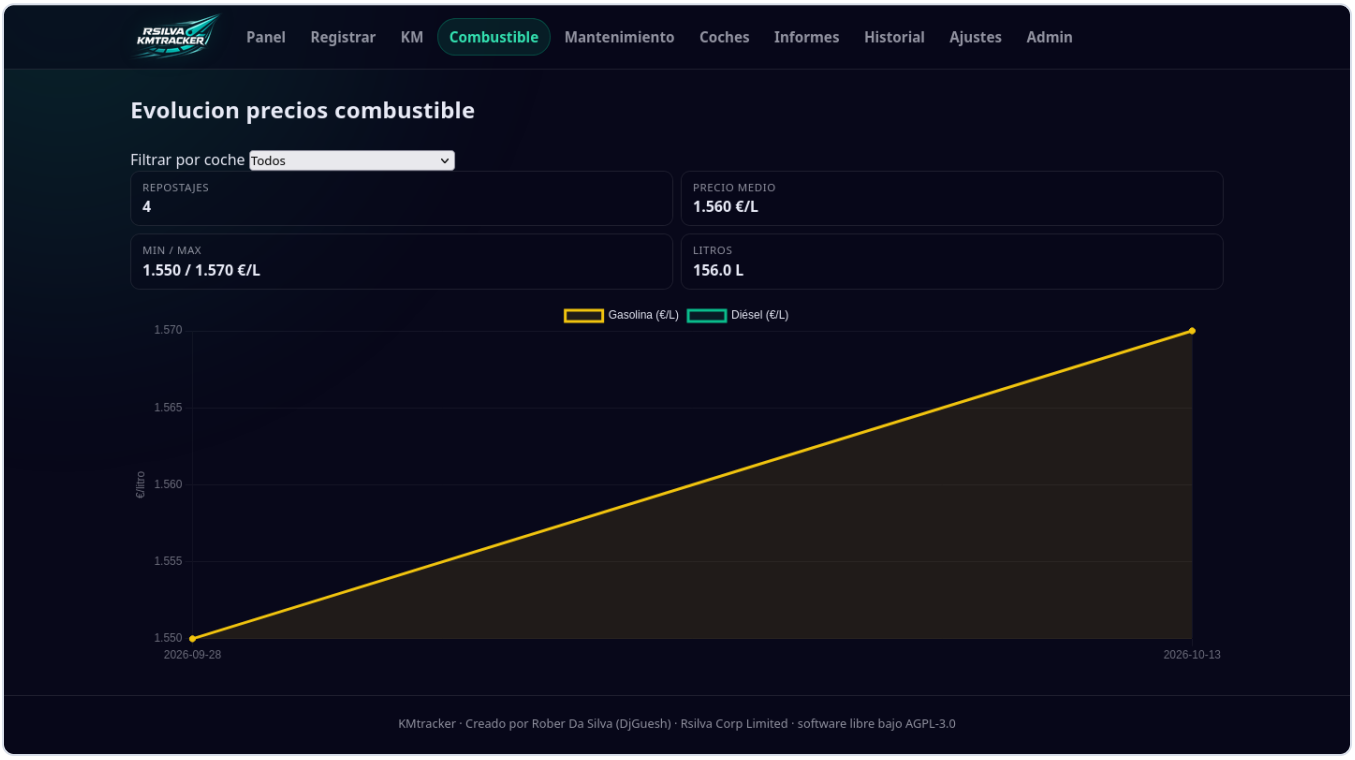
For each refuelling the app stores:

- **Litros** — litres refuelled.
- **Precio por litro** and **total** — price per litre and the amount.
- **Kilómetros** — the odometer reading at that moment.
- **Tipo** — petrol or diesel.
- **Depósito lleno** — tick this when you fill the tank completely.

> **Tip:** for the consumption calculation to be exact, tick **"full tank"** only when you have really filled it up. Between two full tanks, the difference in kilometres and the litres refuelled give the real consumption in **L/100 km**.

Price history

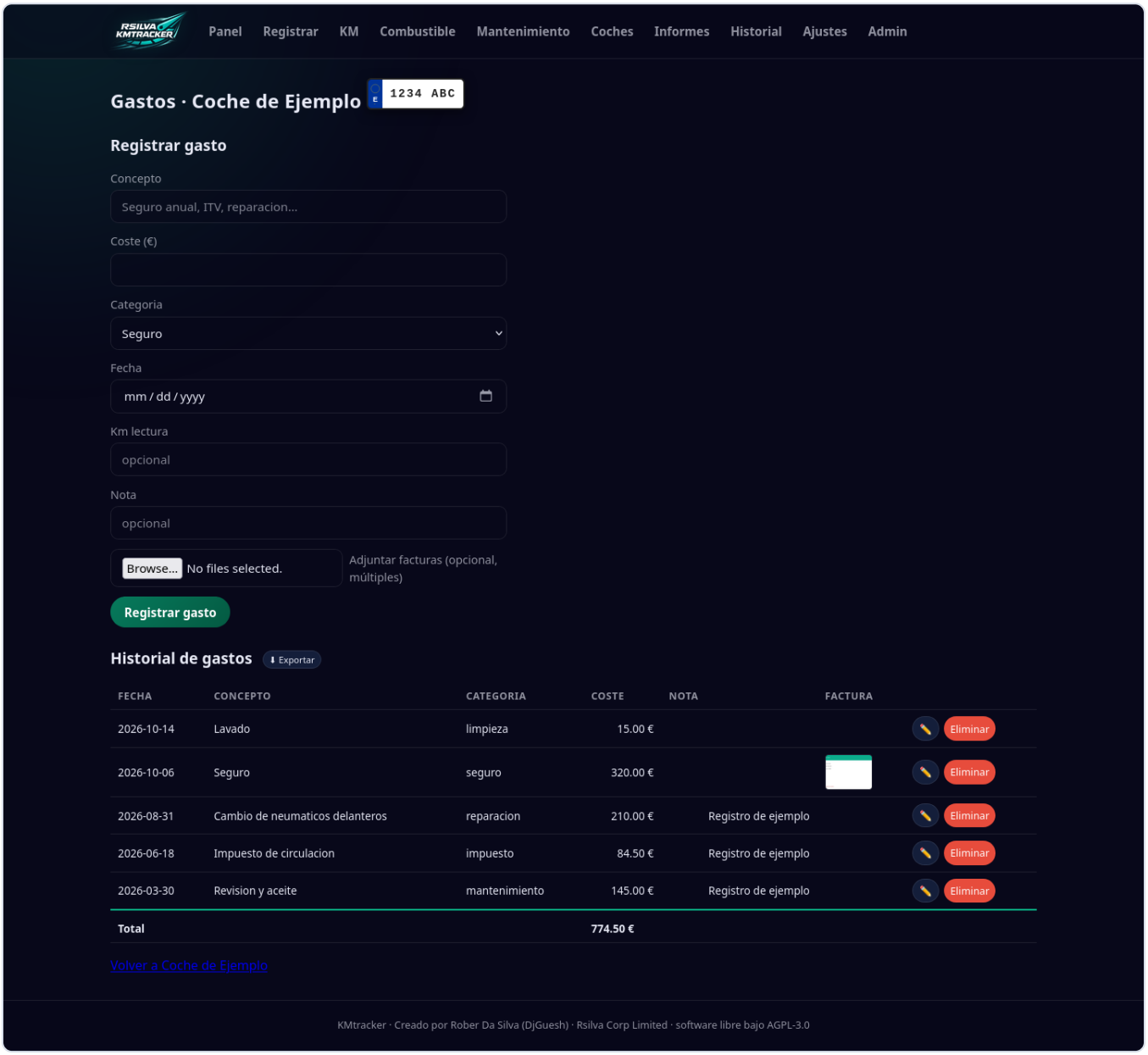
The **Precios** link opens a fuel price history, useful to see how the price has evolved over time.



If needed, you can **edit** a refuelling to correct the odometer or any other value.

7. Expenses

On the vehicle detail page, the **Gastos** (Expenses) section records everything that is not fuel or maintenance.



Each expense has:

- **Date**
- **Concept** — free description (for example, "Annual insurance").
- **Category** — one of: **seguro · ITV · impuesto · reparación · limpieza · mantenimiento · mejoras · otro** (insurance · roadworthiness · tax · repair · cleaning · maintenance · upgrades · other).
- **Cost**
- **Invoice** (optional) — a **photo** of the receipt or invoice.
- **Note** (optional)

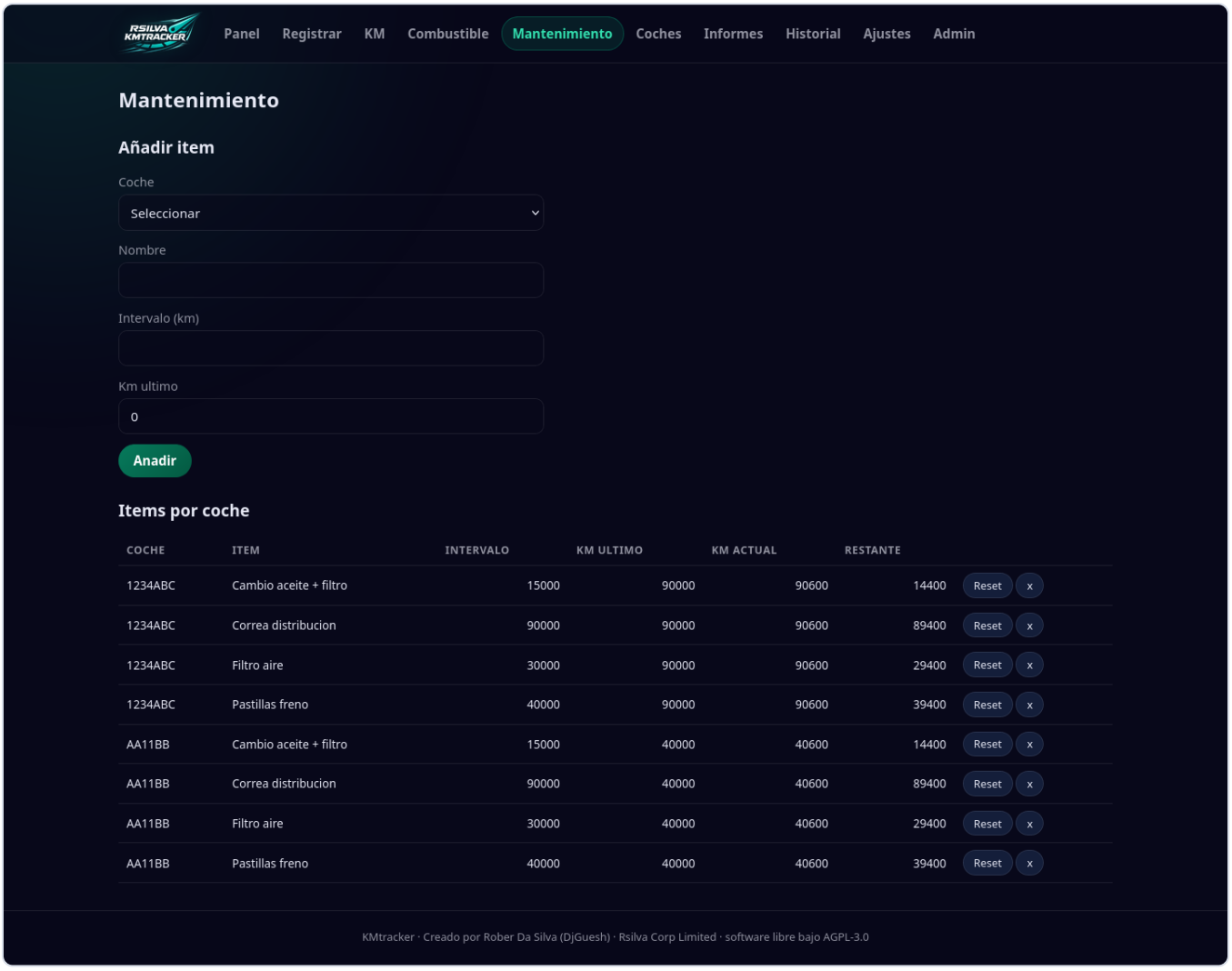
Expenses are totalled per vehicle and per period, and are taken into account in all the dashboard and report statistics.

8. Maintenance

KMtracker distinguishes between **tasks** (what has to be done every so many kilometres) and **services performed** (what you have already done).

Maintenance tasks

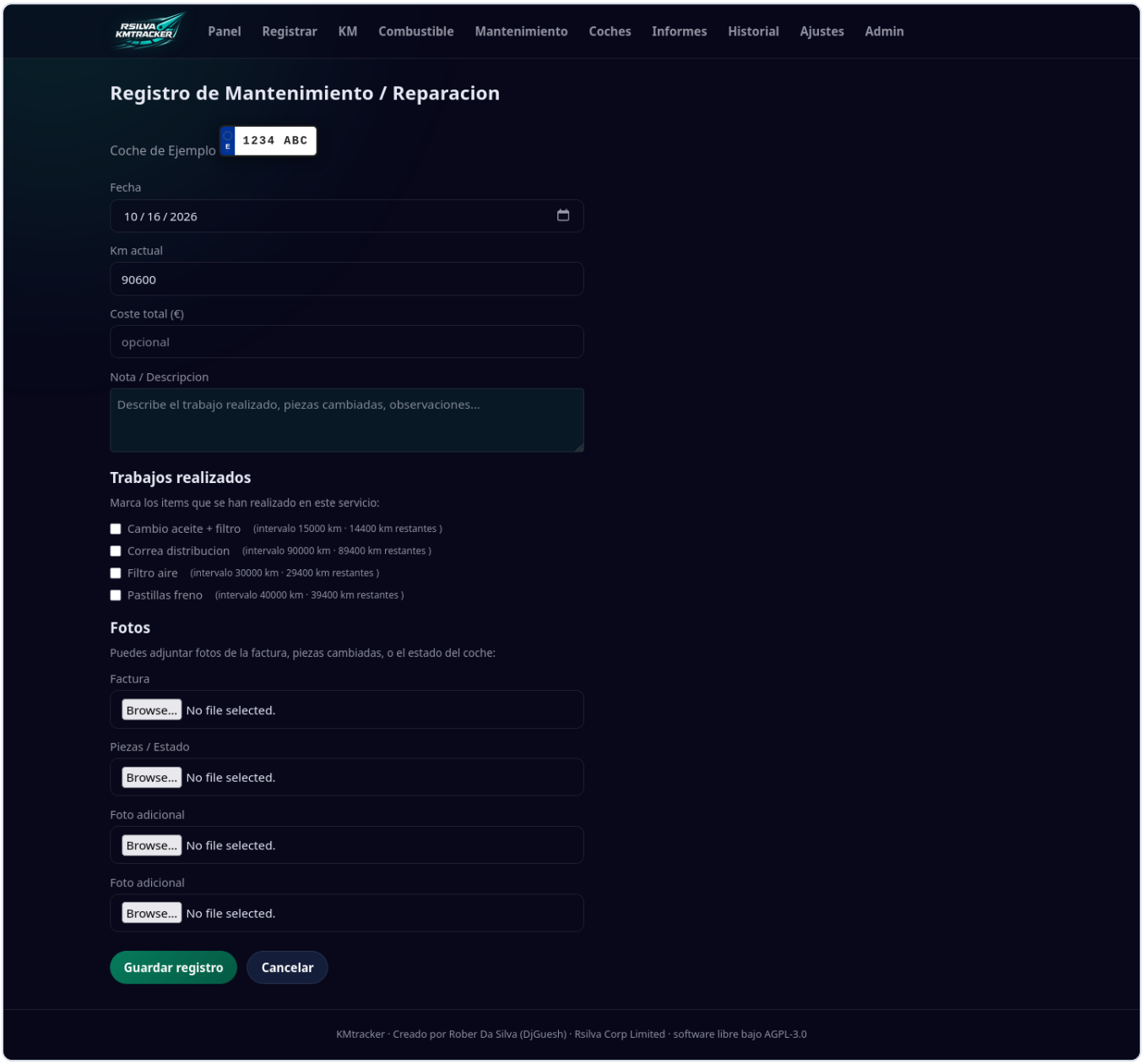
In **Mantenimiento** (Maintenance) you manage the list of control tasks and their intervals in kilometres.



- Each task has an **interval** (for example, "oil every 15,000 km").
- The application shows how many kilometres are **remaining** for each task and warns you when it is **due soon** or **overdue**.

Recording a service performed

To log a maintenance job done, open the vehicle and click **Nuevo** in the maintenance section.



You record:

- The **date** and the **kilometres** at which it was done.
- The **cost** and a **note**.
- The **tasks** covered (ticking them from the list), so their countdown restarts.
- Optionally, **photos**.

Thermal ticket (58 mm)

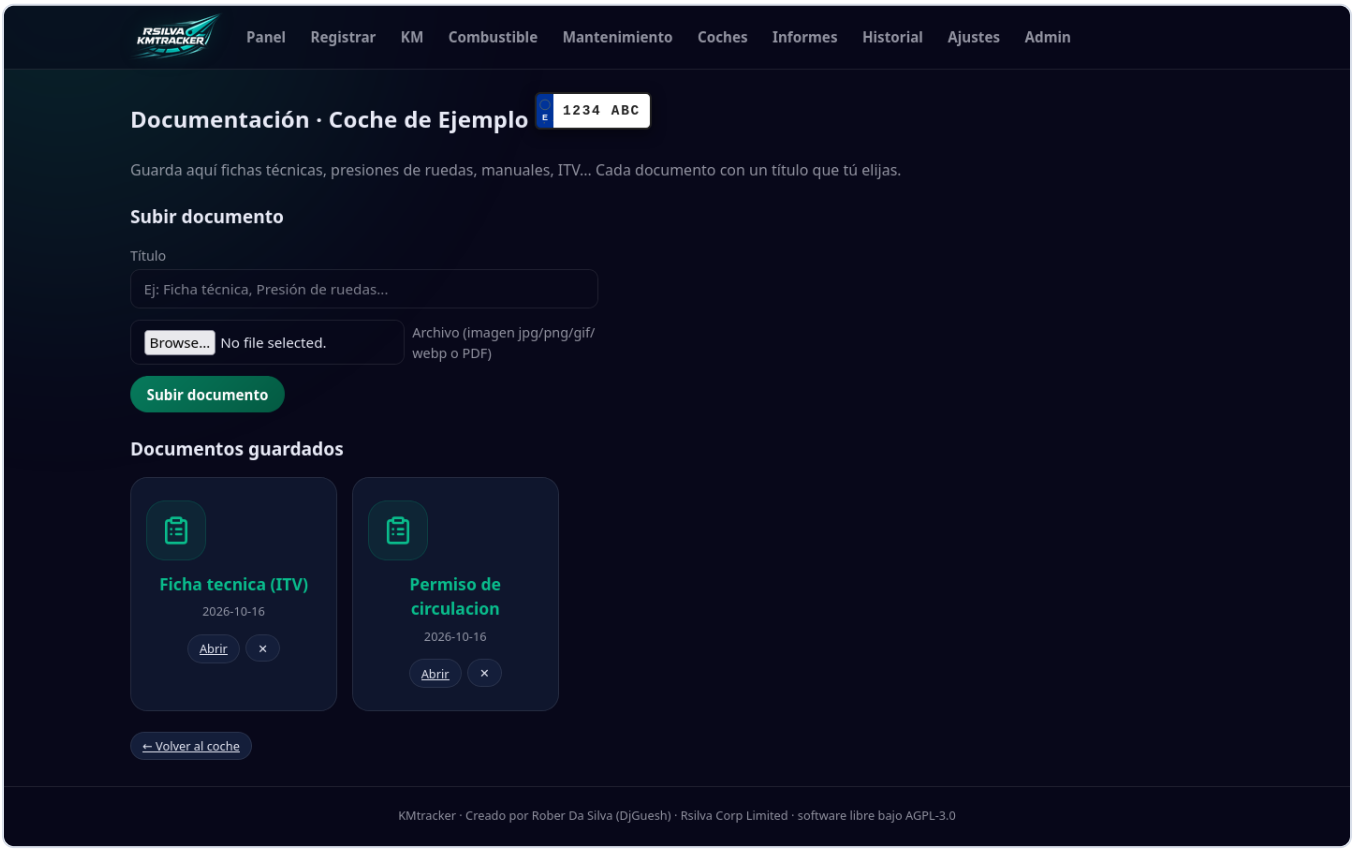
Each service performed can be printed as a **58 mm thermal ticket**, ideal for a receipt printer. The ticket page also lets you save it as a **PNG**.



You can also generate a **PDF** with the maintenance summary of each vehicle from its detail page (the **PDF** button).

9. Documents

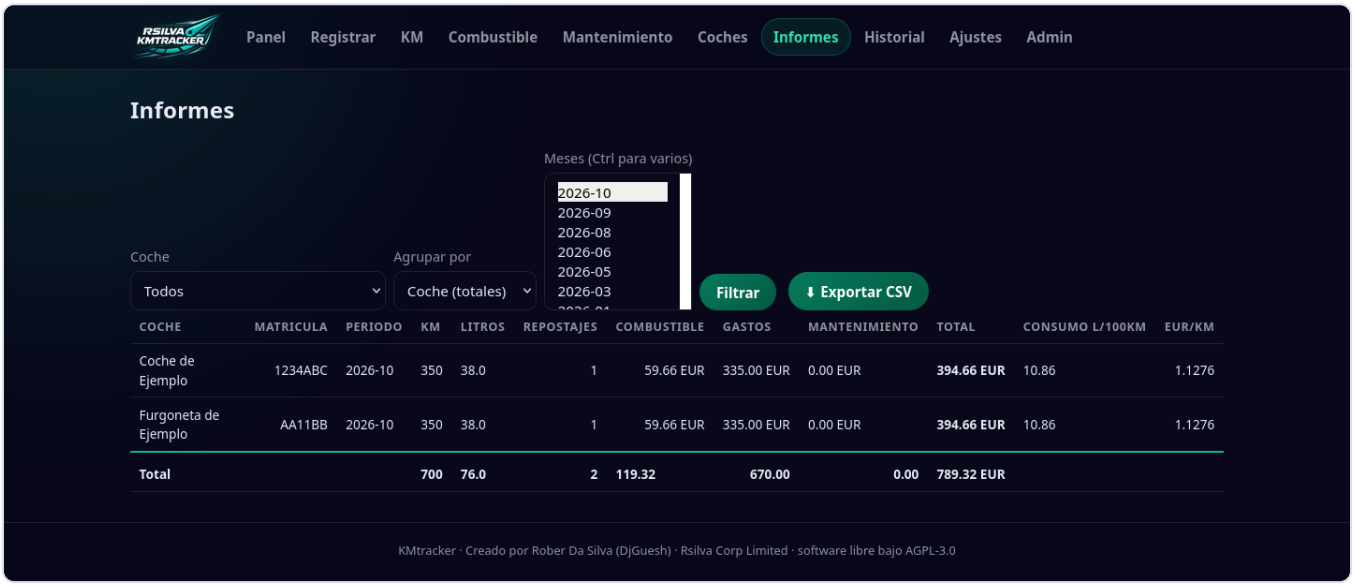
In the **Docs** section of each vehicle's detail page you can store its documentation: registration certificate, technical data sheet, insurance policies, contracts, etc.



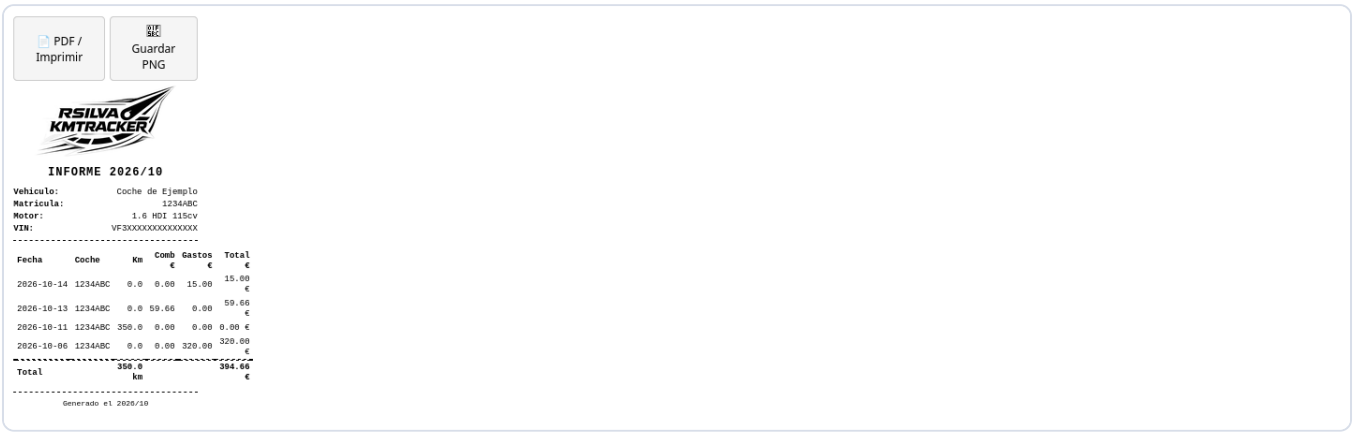
Documents are stored **inside the database**, so they always travel with your backups. You can **download** them whenever you need.

10. Reports

In **Informes** (Reports) you get summaries by month range and by vehicle.



- Choose the **month range** and, optionally, a specific **vehicle** (or all).
- It shows the total kilometres, fuel, expenses and maintenance for the period.
- You can **export to CSV**, a format that opens directly in Excel or any spreadsheet.
- There is a **daily report** of movements and a **report ticket** ready to print.

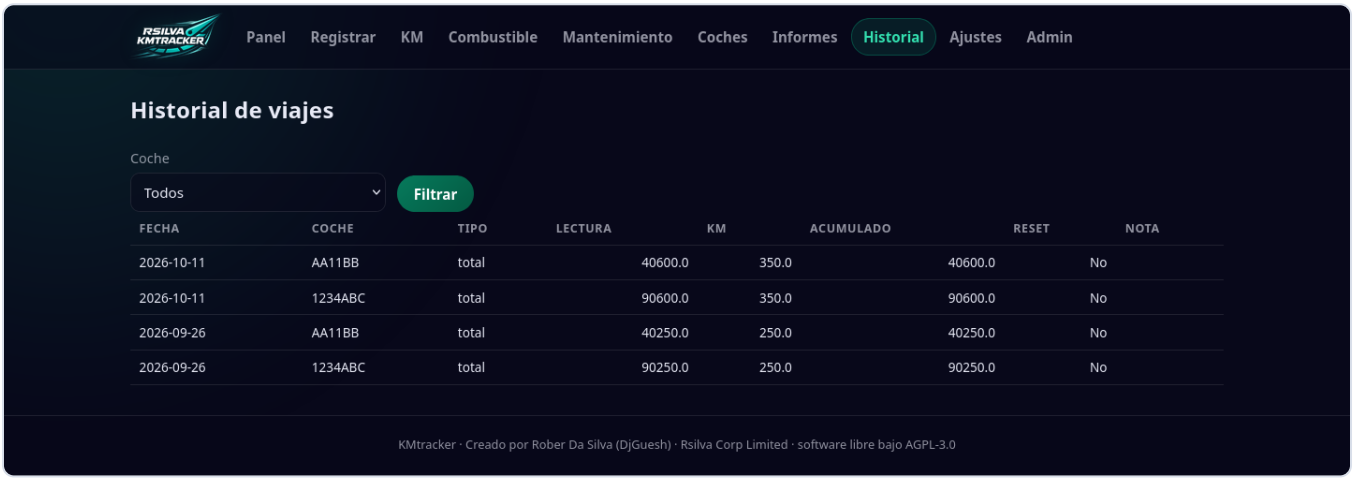


The CSV is also available from each vehicle's detail page, to export only its data.

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11. History

In **Historial** (History) you have the chronological record of activity.

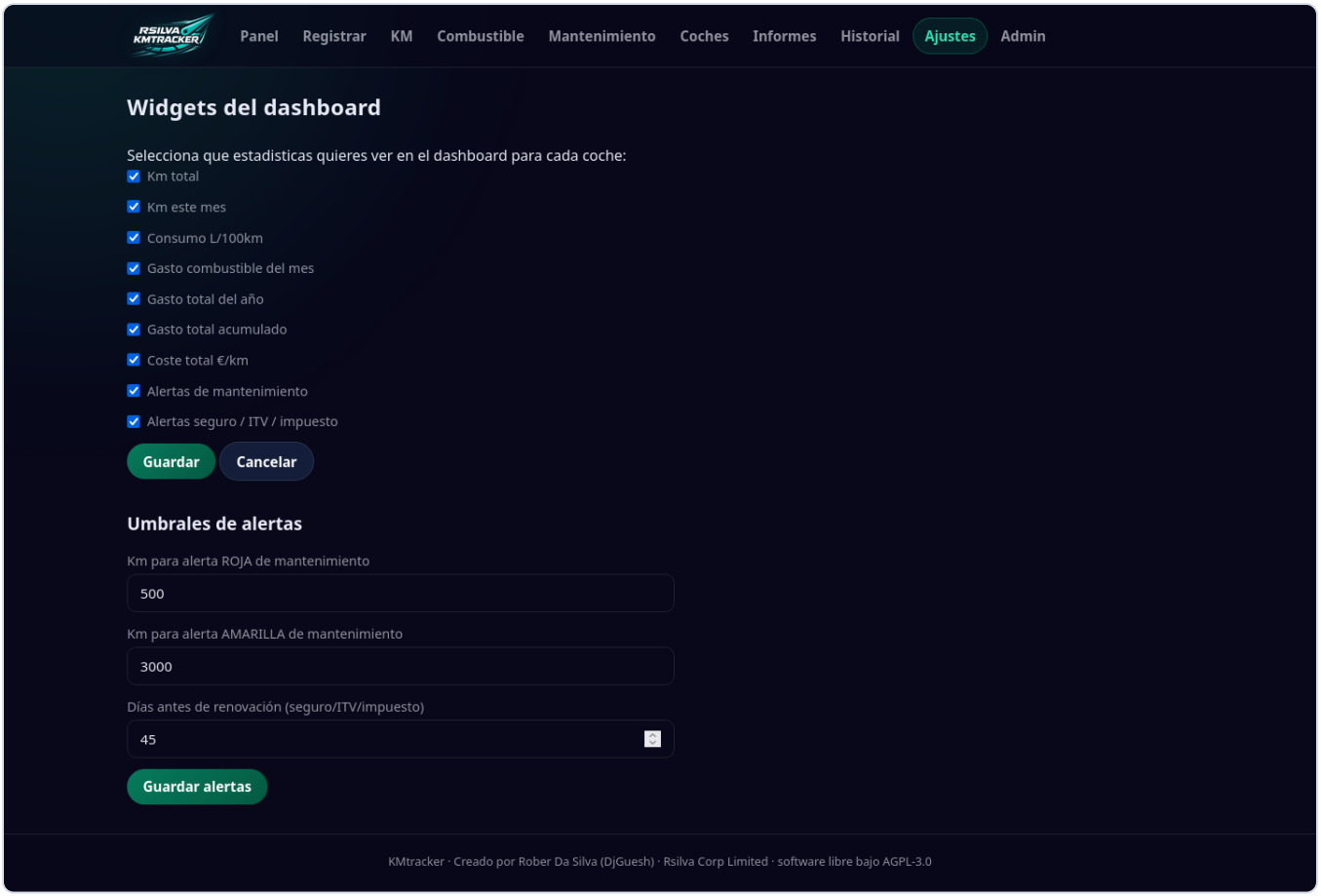


It is the most convenient view to review "what happened" for a vehicle or for the whole fleet: mileage readings, refuellings and other movements, in order.

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12. Settings

In **Ajustes** (Settings) you personalise the dashboard and the warnings.



Dashboard widgets

Tick or untick the statistics you want to see on the dashboard cards:

- Km total · Km this month · Consumption L/100 km · Monthly fuel spend · Total yearly spend · Total cost €/km · Accumulated total spend · Maintenance alerts · Insurance / roadworthiness / tax alerts.

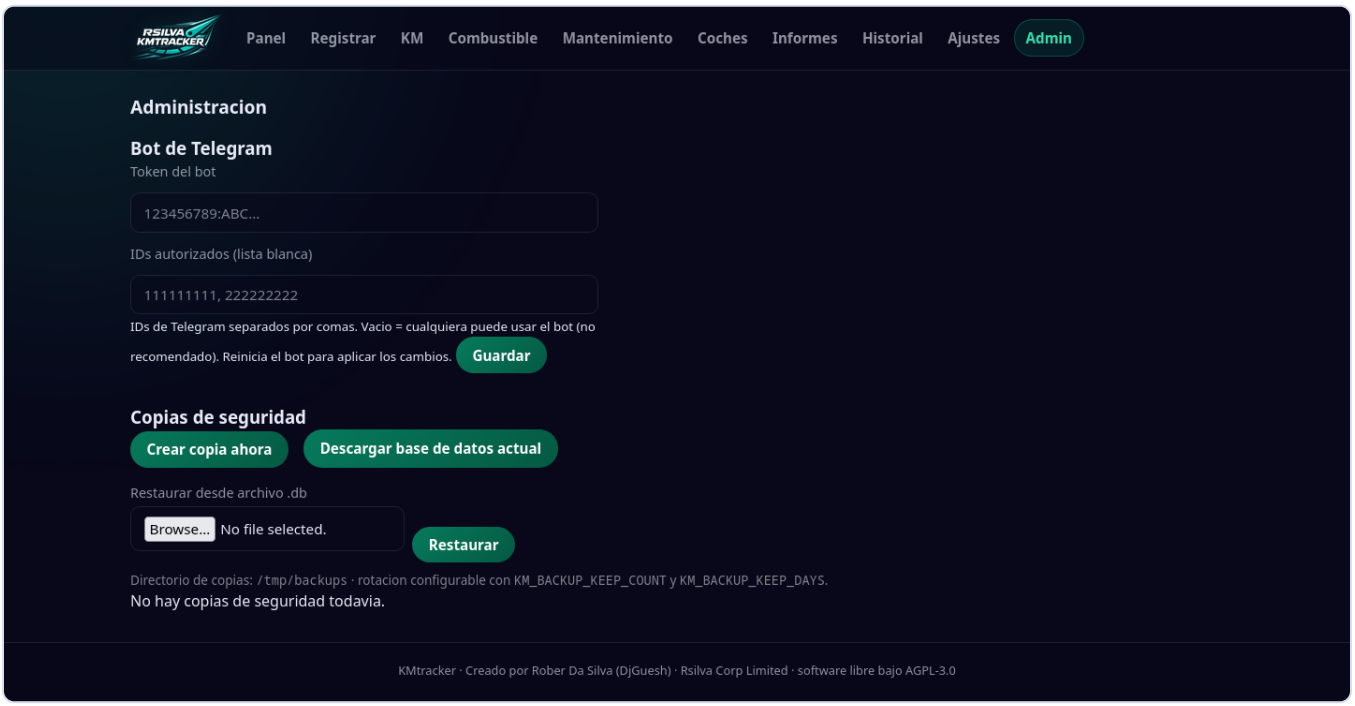
Alert thresholds

- **Km for RED alert** of maintenance (default, 500 km): when a task has fewer than those kilometres left, it is shown in red.
- **Km for YELLOW alert** of maintenance (default, 3,000 km): advance warning.
- **Days before renewal** (default, 45): how far in advance you want to be warned about **insurance, roadworthiness test and tax**.

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13. Administration

In **Admin** you configure the bot and manage backups.



- **Telegram bot** — the bot token and the list of authorised IDs (see the next section).
- **Backups** — create a backup now, list existing ones, download, **restore** or delete them, and download/upload the database file directly.

> These functions are sensitive. Keep the access password safe.

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14. The Telegram bot

The **Telegram bot** is optional and lets you log entries from your phone without opening the browser.

Setup

1. Talk to @BotFather in Telegram, create a bot and copy the **token** it gives you.
2. Find your numeric **Telegram ID** (for example, with @userinfobot).
3. Enter the token and your ID in the **Admin** section of KMtracker (or in the `.env` file, see Installation and configuration).
4. Restart the bot to apply the changes.

> **Very important:** always configure the **allow-list** of authorised IDs. Without it, anyone who finds your bot could read and modify your data.

Available commands

Command	What it's for
`/start`	Welcome and command list.
`/menu`	Menu with buttons.
`/km`	Log a trip.
`/repostar`	Log a refuelling.
`/gasto`	Log an expense.
`/mantenimiento`	Manage maintenance tasks.
`/check`	Summary of a car.
`/alertas`	Alerts for all cars.
`/cancel`	Cancel the current operation.

15. Backups

Your data lives in a single file (the SQLite database), so making backups is very simple.

From **Admin** you can:

- **Create a backup now** (saved as `kmtracker_<date>.tar.gz`).
- **List**, **download**, **restore** and **delete** existing backups.
- **Download** or **upload** the `.db` database file directly.

Backups are created and **purged automatically** by age, with two settings:

- `KM_BACKUP_KEEP_COUNT` — how many backups are always kept (default 10).
- `KM_BACKUP_KEEP_DAYS` — backups older than this many days are deleted (default 30).

> **Tip:** point the backup folder (`KM_BACKUP_DIR`) to a **mounted NAS** or a **cloud-synced folder**, and you'll have off-server copies for free.

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16. Installation and configuration

KMtracker can run in two ways:

With Docker (recommended)

Requirements: **Docker** and **Docker Compose**.

```
git clone https://github.com/Robertomola18/kmtracker.git
cd kmtracker
cp .env.example .env          # edit .env and set KMTRACKER_PASSWORD
docker compose up -d --build
```

Open `http://localhost:5050` and sign in with the password you set.

Without Docker (Python)

```
python3 -m venv venv && . venv/bin/activate
pip install -r requirements.txt
KMTRACKER_DB_PATH=./kmtracker.db KMTRACKER_PASSWORD=your-password \
  python3 app.py
```

Configuration variables (`.env`)

Variable	Default	What it does
`KMTRACKER_PASSWORD`	—	Web access password. **Required** the first time.
`KMTRACKER_DB_PATH`	`/data/kmtracker.db`	Path of the SQLite database file.
`KMTRACKER_SECRET_KEY`	generated in `/data/.secret_key`	Flask session key.
`TZ`	`Europe/Madrid`	Time zone.
`KM_BACKUP_DIR`	DB folder + `/backups`	Folder where backups are stored.
`KM_BACKUP_KEEP_COUNT`	`10`	Number of backups always kept.

Variable	Default	What it does
`KM_BACKUP_KEEP_DAYS`	`30`	Maximum age (days) of backups.
`TELEGRAM_TOKEN`	—	Telegram bot token (also in **Admin**).
`TELEGRAM_ALLOWED_USER_IDS`	—	Authorised Telegram IDs, comma-separated.

Sample data (optional)

Want to try the application with made-up data?

```
docker compose exec kmtracker python3 /app/seed.py
```

Updating

```
git pull
docker compose up -d --build
```

Your data is preserved in the `kmtracker_data` volume.

> **Security advice:** do not expose KMtracker directly to the Internet. Put it behind a reverse proxy with **HTTPS** (Caddy, nginx, Traefik) or a **VPN** such as Tailscale or WireGuard, and publish the port **only on your local network**.

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17. FAQ and troubleshooting

Do I need Internet to use it? No. You only need it for the Telegram bot (optional). Everything else works on your local network.

Where is my data stored? In a single SQLite file inside the `kmtracker_data` volume. You can copy it and take it with you whenever you want.

Can I use it with several people? Access is via a shared password. It is designed for personal or family use, not multi-user with accounts.

What if I record the wrong mileage? You can edit any reading (pencil icon). KMtracker automatically recalculates the affected distances and consumption figures.

Does it run on a Raspberry Pi? Yes. It is designed for modest hardware: Flask + SQLite use very little.

I forgot my password. Set `KMTRACKER_PASSWORD` again in the `.env` file (or environment variable) and restart the application.

The Telegram bot does not respond. Check that the token is correct, that the bot is running and that your Telegram ID is on the allow-list. Restart the bot after changing the configuration.

How do I report a bug or request a feature? Open a GitHub issue. For **security** issues, use a private Security Advisory.

Contact and licence

- **Questions and suggestions:** GitHub Discussions.
- **Bugs and features:** issues.

This project is distributed under the **GNU AGPL-3.0** licence (see LICENSE).



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